

Predicting When Collection Statistics Help Top- k Selection: From Retrieval to Attention and Expert Routing

Team ANLP Doomsday

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Abstract

Retrieval, long-context attention and sparse mixture-of-experts all pick a few items out of many, which documents to return, which cached tokens to attend to, which experts to route through, by an inner product against learned features and a hard top- k . Retrieval learned long ago to down-weight an item that fires across the whole corpus, the collection-frequency term in BM25. On sparse-autoencoder features BM25 reaches 0.582 nDCG@10 where the inner product reaches 0.381, and on SPLADE the order reverses, 0.455 against 0.568. The published account is that the correction helps while a vocabulary stays Zipfian, but it has never been made predictive, so for a new representation the only way to choose a scorer is to run both. We predict the choice in advance. From one frozen encoder we build sixteen representations, eight bases at two granularities, and score each with a single BM25-style function whose four components switch independently, so the inner product and BM25 are two of its configurations. Across fourteen collections, four of them Indic or code-mixed, we regress six diagnostics that cost one forward pass on the gap between the two configurations, 224 rows in all, and validate on held-out basis families and corpora. Attention and routing pose the same choice and reinvent only fragments of the correction, so the fitted predictor applies there without refitting, on one effect-size scale. Whether it holds there is open. An attention sink fires constantly yet cannot be removed, and the strongest experts need not fire often.

1 Motivation and Research Gap

A learned sparse retriever scores a query-document pair as $\langle u_q, u_d \rangle$ over a concept vocabulary. A block-sparse attention kernel estimates how critical KV page i is for query q_t as $r_{t,i} = \langle q_t, c_i \rangle$ over page summaries, then attends to the top- k pages only (Tang et al., 2024). A mixture-of-experts router

gates on $g_e = \langle h_t, w_e \rangle$ and sends the token to its top- k experts (Shazeer et al., 2017; Fedus et al., 2022). These belong to three different literatures and perform one computation, an inner product against a learned non-negative basis followed by a hard top- k .

None of the three accounts for how often a given feature is selected. Classical retrieval does, with a correction in three parts, namely inverse document frequency, which discounts features that fire everywhere, term-frequency saturation, so that repeated evidence stops accumulating linearly, and length normalisation (Robertson and Zaragoza, 2009). Parts of it have been reinvented inside the transformer under other names. Loss-free load balancing adds a per-expert bias driven by recent load (Wang et al., 2024), an additive frequency discount applied during training with no saturation or length term, which replaces an auxiliary loss that pushes toward uniform expert use (Zoph et al., 2022). Streaming inference hand-codes an exception for the most-attended keys in the cache (Xiao et al., 2024). Each is a local fix.

The shortcoming in the published work is narrower than those local fixes suggest. Three papers have run the comparison, all in retrieval, and it favours the correction twice (Clavié et al., 2026; Park et al., 2025) and comes out level once (Han et al., 2026). Clavié et al. and Han et al. explain it the same way, that BM25 helps while a vocabulary keeps its Zipfian structure. None of the three varies the vocabulary systematically, so the account rests on a handful of vocabularies. No one has turned it into a quantity that can be computed before the experiment, or checked it where the same computation runs inside a model.

2 Contributions

1. A controlled family of sixteen representations from one frozen encoder, so basis construction

and granularity vary while representation learning stays fixed. No published comparison varies them systematically.

2. One scoring function with four independent switches, so the plain inner product and BM25 are two configurations of one function, and any gain can be traced to a single mechanism.
3. A predictor for the sign and size of the gain. Six diagnostics are regressed on the gap between those two configurations over 224 rows and validated on held-out basis families and held-out corpora, which turns the Zipfian account into something usable in advance.
4. A test outside retrieval, which no single-setting study can run. The fitted predictor is applied without refitting to KV-page selection and expert routing, on one effect-size scale across the three settings.

3 Background

BM25 scores a document d against query Q as

$$\sum_{i \in Q} \text{idf}_i \cdot \frac{f_i(d) (k_1 + 1)}{f_i(d) + k_1 \left(1 - b + b \frac{|d|}{dl}\right)}, \quad (1)$$

with $\text{idf}_i = \ln\left(1 + \frac{N - df_i + 0.5}{df_i + 0.5}\right)$, the smoothed variant, which stays non-negative. None of these ingredients requires words. Each needs only a representation with non-negative coordinates, a meaningful per-dimension collection frequency df_i , and a magnitude that admits a notion of length. Dense embeddings have none of these, since their coordinates are signed and every dimension is active in every document, which is why dense retrieval is scored with a plain inner product.

Learned sparse features change the picture. Sparse-autoencoder activations, k -means codes and quantised codebooks are non-negative, sparse and count-like. KV page summaries are built from signed keys, but the quantity IDF needs is a selection count, which is non-negative for pages and experts alike and well defined over a calibration set. All three mechanisms are eligible for the same correction, so whether it helps is an empirical question.

4 Related Work

Learned sparse retrieval and its vocabularies. SPLADE learns term weights over the subword vocabulary and keeps the inverted index (Formal et al., 2021; Lassance et al., 2024). Recent work replaces

that vocabulary with a learned one. CL-SR indexes SAE-decomposed dense embeddings as concept units (Park et al., 2025), SPLARE trains an SAE-based sparse retriever directly (Formal et al., 2026), and SAE-SPLADE swaps SPLADE’s MLM head for an SAE encoder (Zong et al., 2026). SPLARE and SAE-SPLADE score by inner product. CL-SR weights activations by IDF instead, and dropping that weighting costs it 11 to 18 percent of its nDCG@10 on TREC-DL, the one published ablation of the correction we study.

Scoring learned sparse features with BM25.

Clavié et al. (2026) show that SAE features taken from a frozen dense retriever have roughly Zipfian collection statistics and are directly BM25-scorable. Their Appendix C holds the sharpest evidence for our question. On one suite BM25 reaches 0.582 nDCG@10 against 0.381 for the inner product on SAE latent terms, and 0.455 against 0.568 on SPLADE-v3, so the two scorers swap places when the vocabulary changes. Han et al. (2026) run the comparison on image features and find (their §4.4) that the two differ by about two points of R@1 on most of their benchmarks, attributing the parity to top- k filtering already suppressing what IDF would down-weight. The same section reports that raising the per-patch activation budget from $k=16$ to $k=128$ makes document frequencies converge and drops R@1 from 0.508 to 0.038, so the effect is distributional. Both papers give the Zipfian explanation of Section 1, which is already published. We take it as given and turn it into a predictor.

Selection inside the transformer. Quest scores KV pages by an inner product against page min-max key summaries and loads only the top- k (Tang et al., 2024). H2O keeps a mix of recent and heavy-hitter tokens (Zhang et al., 2023b), and StreamingLLM preserves the earliest keys as attention sinks (Xiao et al., 2024). Expected Attention ranks a cached pair by how strongly future queries are predicted to attend to it and prunes the low scorers (Devoto et al., 2025). None of these uses the rate at which a page is selected across queries. On the routing side, Su et al. (2026) identify a handful of experts by activation magnitude, a quantity that selection rate does not track, and pruning three of 6,144 degrades the model badly. To our knowledge no work treats retrieval scoring, KV selection and routing as one scoring problem.

5 Proposed Methodology

A modular scorer. For every setting we implement one scoring function in which soft IDF weighting, term-frequency saturation, length normalisation and top- k sparsification are separate switches. Top- k off leaves the basis at its own sparsity, and on it truncates each vector to its k largest coordinates. The uncorrected inner product is the all-off configuration and BM25 is the all-on configuration. The quantity we predict throughout is the margin,

$$m = \frac{Q_{\text{on}} - Q_{\text{off}}}{s(S, c)}, \quad (2)$$

the change in quality between two configurations at matched budget, divided by the standard deviation $s(S, c)$ of that quality across the fixed set of switch configurations in setting S on collection c . The margin is then dimensionless and means the same thing in every setting, which lets a model fitted in one be scored in another. The primary metric is nDCG@10 in Setting A, RULER accuracy in Setting B and MMLU accuracy in Setting C, and any perplexity enters as its negative, so every metric points the same way.

Setting A: retrieval, used for calibration. We build eight concept bases from a single frozen encoder, spanning near-dense to strongly lexical, namely a split-rectified identity map, random projection, PCA, k -means codes, product quantisation (Jégou et al., 2011), residual quantisation, a Top- K sparse autoencoder (Gao et al., 2025), and SPLADE (Formal et al., 2021; Lassance et al., 2024). SPLADE is in the set on purpose, since the reversal we explain is SPLADE against SAE. The eight fall into four families, linear, clustered, quantised and learned, which are the groups the folds remove. Each basis is instantiated at pooled and token-level granularity, giving sixteen representations, with granularity as an explicit variable.

Setting B: KV-page selection. We start from Quest-style page criticality (Tang et al., 2024) and add an online statistic sf_i , the fraction of prefill queries for which page i enters the top- k . The corrected score multiplies the saturated criticality by $\text{idf}(\text{sf}_i)$ and by a length term in $\|c_i\|$, which is the scorer of Eq. (1) with pages for documents. The method needs no training and drops into an existing kernel. Whether attention sinks are exempted is itself an ablation, and if IDF suppresses them and accuracy falls, that is the mechanism result.

Setting C: expert routing. The same correction applies to router logits at inference time on OLMoE-1B-7B (Muennighoff et al., 2025), which activates 8 of 64 experts per layer and whose 6.9B parameters fit one node at 4-bit, with expert selection frequency measured over calibration data as in Setting B. Loss-free balancing (Wang et al., 2024) is the additive, training-time special case we compare against. We also check whether the correction protects or destroys the Super Experts of Su et al. (2026), which sets frequency against magnitude directly.

Diagnostics and the predictor. From a single forward pass over calibration data we compute six measurements, namely active fraction per item, the Zipf exponent of the selection-frequency curve, the Gini coefficient of activation mass, head mass, the variance of IDF values, and sink share, meaning the fraction of score mass held by the top 1% most-frequently-selected features. Han et al. (2026) report several of these descriptively but predict nothing with them. One row of the regression pairs a representation with a collection, because collection frequency belongs to a corpus and a basis therefore has no single frequency curve. Measuring it again on each collection is free, since indexing already performs that pass. Sixteen representations across the fourteen collections of Section 6 give 224 rows for six regressors.

Theory. We aim to characterise when an IDF-weighted linear scorer over a learned basis provably reduces to a bilinear inner product on the dense embeddings, the collapse condition. Where it holds the two scorers must agree, which explains part of the grid instead of tabulating it.

6 Datasets

Setting A runs over fourteen collections drawn from twelve corpora. Bases are fitted and hyperparameters tuned on MS MARCO passage (Bajaj et al., 2016), 8.8M passages, and nowhere else, so every other collection tests generalisation. LIMIT (Weller et al., 2026) is built from a capacity argument, that the top- k sets a query can return are bounded by embedding dimension, and lexical scoring does well on it because its dimension is high, which is the axis our bases are built to span. TREC-DL 19 and 20 and six BEIR datasets (Thakur et al., 2021) cover the domains over which the published reversal was reported. The two TREC-DL query

sets run over the MS MARCO documents, so those three collections share one set of diagnostics.

Because collection frequency is a property of a language and a corpus rather than of a model, the remaining four collections are the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a) and a set of Hindi-English code-mixed queries, where transliteration flattens the document-frequency curve. SPLARE reports multilingual retrieval quality on Indic languages but measures no frequency structure at all, and the multilingual evaluation in Zong et al. (2026) covers six MIRACL languages of which none is Indian. Corpus-level measures of code-mixing already tell such datasets apart where token-level ones cannot (Kodali et al., 2022). Because every collection contributes its own rows, these four enter the fit itself.

Setting B uses RULER (Hsieh et al., 2024), needle-in-a-haystack and LongBench on Llama-3.2-1B/3B-Instruct and Qwen2.5-7B-Instruct. Setting C uses OLMoE-1B-7B with WikiText for perplexity and MMLU and GSM8K for downstream accuracy.

7 Evaluation Plan

For Setting A we report nDCG@10, MRR@10 and Recall@100/1000 alongside index size, FLOPs and latency, since a gain that destroys the efficiency of sparsity is worth less. As a pipeline check we reproduce the Appendix C reversal of Clavié et al. (2026) within a stated tolerance before trusting our own numbers. Setting B reports accuracy at matched token budget against Quest and H2O (Tang et al., 2024; Zhang et al., 2023b).

The headline result is the diagnostics-to-margin regression. The fitted model emits three quantities we can score, the sign of the margin, measured by accuracy and AUC, its magnitude in units of s , measured by held-out R^2 in retrieval and by mean absolute error when it is carried to Settings B and C without refitting, and the ordering of configurations within a setting, measured by Kendall’s τ . Rows are nested by representation and by corpus, so we cluster standard errors on both axes and validate by holding out whole groups on each, the four basis families and the twelve corpora, reported separately, since generalising to an unseen construction method and generalising to an unseen corpus are different claims. Grouping is by corpus rather than by collection, because query sets over one document set share their diagnostics exactly.

Since much of the raw variance sits between corpora, the number we report is the gain in held-out R^2 over a baseline carrying corpus indicators alone, which no model wins by learning which corpus is easy.

The analysis is fixed in advance, since the grid produces several hundred pairwise comparisons. The primary comparison is all-on against all-off per representation and component ablations are secondary. Significance uses a paired bootstrap over queries with Holm–Bonferroni correction within each family of comparisons, effect sizes are reported alongside p -values, and rows are weighted by the inverse variance of the margin estimate, since a small code-mixed query set gives a noisier margin than MS MARCO dev. The transfer holds if the predicted sign is correct on at least 70% of held-out rows in Settings B and C and the mean absolute error there is no worse than 1.5 times the residual spread inside retrieval.

8 Schedule

Nine weeks separate this proposal from the final submission. The retrieval grid runs first and Setting B follows it, so a transformer result exists by the mid submission, leaving the predictor, the transfer and the ablations for the weeks after that. The four of us split the work by setting, with the scorer and the diagnostics shared.

9 Risks and Limitations

The failures are the informative outcomes. Confirming that BM25 helps on Zipfian vocabularies only repeats what the cited papers predict. A predicted gain where the measured effect is negative would show contribution to the output and discrimination between queries coming apart inside the transformer. Diagnostics promising a gain that never appears would falsify the Zipfian account as stated. If the predictor is weak on both held-out axes we report the grid as a benchmark and the collapse condition as the theoretical contribution. If Setting C proves infeasible we fall back to Settings A and B, which still support the transfer claim.

Three limitations stand out. Setting A uses a single frozen encoder, so basis and encoder effects are not fully separable. Setting C is inference-time only, so we claim nothing about training dynamics. The diagnostics are correlational throughout, and only the transfer test carries any causal weight.

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